THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 11: **Earth, Moon and the Sun** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The first quarter moon is seen as: 1

(a) full
(b) crescent
(c) half disc, right half lit (northern hemisphere)
(d) nothing

2. During a total lunar eclipse the Moon appears: 1

(a) green **(b)** blue **(c)** dark reddish **(d)** bright white

3. The length of day on the equator throughout the year is about: 1

(a) 12 hours **(b)** 24 hours **(c)** 6 hours **(d)** 18 hours

4. Half the Moon's surface is always lit by the Sun because: 1

(a) it makes light **(b)** stars light it
(c) the Sun always shines on half of it **(d)** the Earth shines on it

5. Which eclipse can be watched safely with the naked eye? 1

(a) lunar **(b)** solar **(c)** both **(d)** none

6. At the North Pole in June: 1

(a) the Sun does not set **(b)** there is no light
(c) day and night are equal **(d)** the Sun does not rise

7. The number of days between a new moon and the next full moon is about: 1

(a) 15 **(b)** 29 **(c)** 30 **(d)** 7

8. The solar eclipse observed on the Earth is visible only: 1

(a) in a narrow region **(b)** at the equator only
(c) on the Moon **(d)** all over Earth

9. At the North Pole in December: 1

(a) day equals night **(b)** the Sun does not rise
(c) sunshine all day **(d)** the Sun does not set

10. The number of days from a new moon to the first quarter is about: 1

(a) 29 **(b)** 15 **(c)** 1 **(d)** 7

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain the term 'lunar month'. 2

12. Explain why the North Pole has six months of day and six of night. 2

13. Why does the Moon appear to rise about 50 minutes later each day? 2

14. Why is a solar eclipse rarer than a lunar eclipse at a given place? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Compare the day length in Delhi in June and December and give the reason. 3

16. Why is the full moon bright but not blinding? 3

17. How does the position of the Sun in the sky change during the year? 3

OR

Explain why Earth experiences four main seasons in temperate regions but only two or three in tropical India. 3

18. Why does a solar eclipse last only a few minutes at a place? 3

OR

Explain why a stick's shadow is shortest at noon. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain why solar eclipses happen at new moon and lunar eclipses at full moon, and why they do not happen every month. 5

OR

Describe how ancient Indian astronomers observed the Sun and Moon and made calendars. 5

20. Describe the experiment to show the cause of seasons using a globe, a lamp and a tilted axis. 5

OR

Explain a pinhole method to observe a solar eclipse with the steps and safety. 5

END OF QUESTION PAPER — Total: 40 Marks

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